WELLSENSE · USER REQUIREMENTS

WellSense user requirements

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	WLS-URS-001	Version	0.1
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/03a-user-requirements.md @ 9c67a3d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade), drawn from the users and stakeholders of WLS-PRB-001 v0.7 (Tables 1 to 3) and the requirements of WLS-REQ-001 v0.8

WellSense user requirements

What the people who share an aquifer, look after a well or run a monitoring programme need from a water level logger, written from their side. The product requirements that meet these needs are in WLS-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix WLS-RTM-001 (`03b-traceability.md`). These needs are the author's reading of the users in WLS-PRB-001 and have not yet been checked in co-design; the first candidate partner, ACWADAM, has not been approached (WLS-DEC-001). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Trustworthy level record	We can see whether the water table is falling and by how much, with readings close enough to a manual tape that we trust them in a meeting	High	Village water committee or farmer group (PRB Table 2)
UR-002	Performance	Our wells, not ideal ones	It works in the boreholes we have: 100 to 200 mm, 20 to 150 m deep, with water from a few metres to more than 50 m down	High	Farmer group, caretaker, ACWADAM (PRB Tables 1 and 3)
UR-003	Performance	Pumping and recovery	We can watch how the level drops while the pump runs and how it recovers, including a pumping test	Medium	Farmer with an irrigation well (PRB Table 2)
UR-004	Logistics	Nothing lost when the network drops	Readings are kept when the radio link fails and sent later	Medium	WASH NGO or agency (PRB Table 2)
UR-005	Deployment and assembly	No extra power	It runs from the lab's shared FieldNode core and its small solar panel, with no grid supply and no extra battery	High	Caretaker (no grid power, PRB Table 2); FieldNode project (PRB Table 3)
UR-006	Safety	Our drinking water stays safe	Nothing lowered into the well can contaminate it, and the wellhead stays sealed against surface water, insects and animals	High	Village water committee, well owner, water authority (PRB Tables 2 and 3)
UR-007	Deployment and assembly	Leave the pump alone	It fits beside the existing pump, riser and cable without touching them, and without pulling the pump where possible	High	Well owner and pump installer (PRB Table 3)
UR-008	Deployment and assembly	Quick to install	Two trained people with hand tools can install it in one visit	Medium	Caretaker, NGO field team (PRB Table 2)
UR-009	Cost	Affordable for a community	The parts cost a small fraction of a commercial logger, so a group can afford one per shared well	High	Village water committee, WASH NGO (PRB Table 2)
UR-010	Usability	Readable on a basic phone	We can read level below ground, daily drawdown and the change against last year on a basic phone, and export the data	Medium	Farmer group, NGO (PRB Table 2)
UR-011	Regulatory	Our data	The community that hosts the well owns the data and decides what to share; only water levels are recorded	Medium	Gram panchayat water committee (PRB Table 3)
UR-012	Environment and durability	Survives the wellhead	It survives sun, heat to 55 °C, monsoon rain and splash at the wellhead for years	High	Caretaker, farmer group (PRB Tables 1 and 2)
UR-013	Safety			Medium	

ID	Category	Title	Need	Priority	Source
		Hard to tamper with	The wellhead parts lock and no cable is exposed at a height people or livestock can reach		Caretaker (tampering and livestock, PRB Table 2)
UR-014	Maintenance	Light upkeep	The caretaker can check and correct it with a tape and change the desiccant once a quarter, with nothing else to service between visits	Medium	Caretaker (PRB Table 2)
UR-015	Safety	Safe to work on	The people installing and checking it are not exposed to dangerous voltages, even beside a mains-powered pump	High	Caretaker, pump installer (PRB Tables 2 and 3)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.